



DATA SCIENCE FIELD NOTES / 02A

BUSINESS UNDERSTANDING REPORT

Loan Approval Data Preparation - defining the problem before modelling

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Educational portfolio report. This work does not constitute a production lending model or lending policy.

Executive summary

This report frames the loan-approval practice project before any predictive model is developed. The supplied dataset records applicant characteristics, requested loan information, credit-history flags and historical approval outcomes. The project asks whether these records can support an educational binary-classification workflow while preserving responsible-use boundaries.

Core question: Can historical application information support a transparent learning exercise that estimates recorded loan-approval outcomes, without presenting the result as a real lending decision system?

Business context

Lending decisions require consistent assessment of risk, affordability and policy requirements. Historical application data can help analysts identify patterns associated with earlier outcomes, but those patterns may also contain undocumented policy choices or social bias. Predictive analysis should therefore support investigation and decision design, not replace responsible lending practice or qualified human oversight.

Problem statement

The educational lender is assumed to receive many applications containing different forms of applicant and loan information. Manual review alone can be slow and inconsistent. The analytical task is to prepare the supplied data and later test whether information available at application time can estimate the historical target, `Loan_Status`.

Decision and intended use

- Primary educational decision: determine whether the available data is sufficiently structured for a later classification experiment.
- Potential demonstration use: produce a probability or class estimate for the historical approval outcome.
- Excluded use: automatically approving, rejecting or ranking real applicants.
- Human role: interpret evidence, review policy and fairness concerns, and challenge model errors before any operational proposal.

Project objectives

- Inspect data structure, completeness, types and duplication.
- Document the target and predictor variables.
- Resolve missing values using transparent educational rules.
- Create interpretable features for total income and a simple loan-income ratio.
- Encode and scale variables where required for later modelling.
- Identify redundant features and initial associations without treating them as causal evidence.

- Publish reproducible outputs and clearly state responsible-use limitations.

Stakeholders

Stakeholder	Interest	Required safeguard
Applicants	Fair and understandable treatment	No automated real-world use; review subgroup errors and harmful proxies
Lending analysts	Consistent evidence and efficient review	Transparent features, documented limitations and human oversight
Risk and compliance teams	Policy, legal and risk alignment	Independent validation, governance and auditability
Data-science team	Reliable and reproducible modelling	Leakage-safe pipelines, versioned data and monitored performance
Project reviewer	Evidence of methodological learning	Clear separation between completed work and future modelling

Analytical approach

The later modelling problem is a supervised binary-classification task. The positive and negative class labels are derived from the historical `Loan_Status` field. The current repository completes the data-preparation phase and uses descriptive statistics, missing-value analysis, correlation and exploratory feature importance to understand the data before a final model is selected.

Element	Definition
Unit of analysis	One loan application
Target	Historical <code>Loan_Status</code> : Y or N
Proposed task	Supervised binary classification
Current phase	Inspection, cleaning, feature engineering and preprocessing
Not yet claimed	Validated predictive performance, deployment or operational lending value

Data requirements

A valid future experiment requires variables that are available at the intended decision time, consistently defined and ethically defensible. The current dataset provides applicant demographics, employment and education categories, income fields, requested loan amount, term, credit-history flag, property area and the historical outcome.

- Application identifiers must not be treated as ordinary predictors.
- Missingness and category definitions must be documented.
- Financial units and collection procedures should be confirmed before business interpretation.
- Sensitive personal variables require necessity, legal and fairness review.

- All transformations used for model evaluation must be fitted on training data only.

Success criteria

Dimension	Provisional criterion
Data quality	Published outputs contain no unexpected missing values, duplicate rows or non-numeric fields in the numeric export
Reproducibility	A clean environment can recreate the processed outputs from the raw CSV
Predictive evaluation	Future work compares against a transparent baseline using class-sensitive metrics, calibration and error analysis
Fairness	Future work evaluates subgroup performance and excludes unjustified sensitive attributes or proxies
Communication	Claims remain limited to the evidence and do not imply production readiness

Expected value

If later validated with suitable data and governance, predictive analysis could help structure risk review, prioritise manual investigation and reveal inconsistencies in historical decisions. In this portfolio project, the immediate value is methodological: creating a transparent, reproducible foundation for responsible model development.

Risks, assumptions and limitations

- The 614-record practice dataset is small and may not represent any actual lending population.
- Historical approval labels can reflect undocumented policy choices or bias.
- The source does not fully document financial units, collection processes or decision timing.
- The loan-income ratio omits expenses, debt, interest rates and repayment capacity.
- Associations and feature importance do not establish causation or legitimate policy relevance.
- A model trained on this dataset must not be used to make real lending decisions.

Conclusion

The project is suitable as an educational classification and preprocessing exercise, provided its limitations remain explicit. The next phase should convert the current transformations into a leakage-safe scikit-learn pipeline, establish a baseline, evaluate performance and calibration, and review fairness before considering any operational scenario.